Ryan P. Seaman

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SUMMARY

Biomedical informatics researcher and MMSc candidate at Harvard Medical School with experience in collaborative data analysis, research visualization, and scientific training. Seeking cross-functional, people-facing product and project roles across biotech, health tech, and technology to translate complex biomedical science into impactful, user-focused solutions.

EXPERIENCE

Graduate Research Assistant, HIDIVE Lab Dec 2025 - Present
Harvard Medical School | Boston, MA

- Developing VitessceGen, a multi-agent AI tool that generates interactive visualization configurations from spatial and single-cell biomedical data.
- Architecting user prompt-to-visualization workflows that translate natural-language researcher queries into tailored, coordinated multi-view visual configurations.

Summer Research Intern, HIDIVE Lab May 2026 - Aug 2026
Harvard Medical School | Boston, MA

- Created the full annotation and narrative system within the existing Vitessce framework, with supervision and guidance from Mark S. Keller and Nils Gehlenborg.
- Connected narrative text, saved views, and visual annotations into guided stories for spatial and single-cell data; manuscript in preparation.

Bioinformatician Jan 2023 - Aug 2025
MDI Biological Laboratory, Comparative Genomic and Data Science Core | Bar Harbor, ME

- Collaborated with research labs on biomedical data analysis and developed tools that help researchers explore their own datasets.
- Built scscape, an open-source Nextflow pipeline for reproducible, multi-sample single-cell RNA sequencing analysis.
- Maintained GCP/AWS infrastructure for research analysis and data storage.

EDUCATION

Harvard Medical School Expected May 2027
MMSc, Biomedical Informatics | Boston, MA

Colby College Jan 2023
BA, Computational Biology | Waterville, ME | Varsity soccer, four years

TEACHING & MENTORING

Harvard Medical School Oct 2026 - Present
Teaching Fellow | Boston, MA

- BMI 706: Data Visualization for Biomedical Applications (Instructor: Dr. Nils Gehlenborg).

MDI Biological Laboratory 2023 - 2025

- Co-taught a three-day single-cell course in 2024; created teaching materials and the course website.
- Taught Python, held office hours, and mentored summer interns on computational research projects.

TECHNICAL SKILLS

Python, R; biomedical data analysis and visualization; single-cell and spatial genomics.

SELECTED PUBLICATION

Seaman RP, et al. (2024). A cloud-based training module for efficient de novo transcriptome assembly using Nextflow and Google Cloud. Briefings in Bioinformatics.